Nordic

U1
nRF54L10
Nordic Semiconductor
NRF54L10-QFAA-R

+3V0 → VDD (10, 22, 36, 47, 48)

AE1
Antenna_Chip

L1 3.5N
L2 3.5N
L3 2.7N

BLE_PCB_ANT — RF — L1 — RF2 — RF — L2 — RF1 — RF — L3 — RF_BLE_ANT_NRF_IC1 — RF — 31 ANT

R9 22K → GND

C3 0.3P → GND
C5 2.0P → GND
C9 1.5P → GND

32 VSS_PA → GND

Note: VSS_PA to GND PAD ONLY!

XC1 → 34 XC1
XC2 → 35 XC2

Y1
32M
CTS-Frequency Controls
416F32022CKR

C2 8P → GND
C6 8P → GND

TP1 RESETB → 30 RESET
TP2 SWD_CLK → 26 SWDCLK
TP3 SWD_IO → 25 SWDIO
TP4
TP5
+3V0 / GND

VSS PAD (44, 49) → GND

P0.00	23	P0.00
P0.01	24	P0.01
P0.02	27	P0.03_PWM
GRTCPWM	28	
P0.04	29	
XL1	1	
XL2	2	
P1.02	3	BT_STAT
P1.03	4	P1.03
P1.04	5	
AIN1	6	AIN1_VBAT
AIN2	7	P1.06_AIN2
AIN3	8	VEXT_DIS_L
CLK16M	9	USB_DRV
P2.00	11	
P2.01	12	
P2.02	13	P2.02_SPI_SDO
P2.03	14	P2.03
P2.04	15	P2.04_SPI_SDI
P2.05	16	P2.05_SPI_CS_L
P2.06	17	P2.06_SPI_SCK
UARTE_RXD	18	NRF_RX
UARTE_TXD	19	NRF_TX
P2.09	20	P2.09
P2.10	21	CLK_CTL
P1.09	37	P1.09
P1.10	38	
P1.11	39	P1.11_I2CSDA_AIN4
P1.12	40	P1.12_I2CSCL_AIN5
P1.13	41	P1.13_TIMR10_AIN6
P1.14	42	P1.14_TIMR10_AIN7

L4 4.7U

DCC (46) — Power — DCC — L5 120R @ 100MHz — C15 2.2U → GND
DECD (45) — Power — DECD
DECRF (33) — Power — DECRF
DECA (43) — DECA

C12 10N → GND
C13 10N → GND
C14 2.2U → GND

Power

+BATT → U2 EA8103T5R
1 RUN
2 GND
3 SWITCH
5 FBK
4 PWR

L6 2.2uH
C18 10uF → GND
C16 22uF → GND

+3V0 → R2 965K → FB → R3 499K → GND

CP2102N-A02-GQFN28R USB-UART CIRCUIT

U4
CP2102N-A02-GQFN28R

+3V0 → R6 10K → 9 RST
+3V0 → 7 VREGIN
VBUS → C19 0.1uF → GND

DCD	1	
RI/CLK	2	
VBUS	8	
D+	4	USB_D_P
D−	5	USB_D_N
CTS	23	
RTS	24	RTS
RXD	25	NRF_TX
TXD	26	NRF_RX
DSR	27	
DTR	28	DTR
NC	10	
VDD	6	+3V0
SUSPEND	11	
SUSPEND	12	
CHREN	13	
CHR0	15	
CHR1	14	
GPIO0_TXT	19	
GPIO1_RXT	18	
GPIO2_RS485	17	
GPIO3_WAKEUP	16	
GPIO4	22	
GPIO5	21	
GPIO6	20	
EXP	29	
GND	3	GND

C21 0.1uF → +3V0 / GND

Sensors

+3V0 → R7 4.7K → P1.11_I2CSDA_AIN4
R8 4.7K → P1.12_I2CSCL_AIN5

IMU

U5
ICM45686

P2.04_SPI_SDI	1	SDO
P2.02_SPI_SDO	14	SDx
P2.06_SPI_SCK	13	SCx
P2.05_SPI_CS_L	12	CSB
P2.03	4	INT1
CLK	9	INT2
	5	VDDIO
	8	VDD
	6	GNDIO
	7	GND
ASDx	2	
ASCx	3	
OCSB	10	
OSDO	11	

+3V0 → C22 0.1uF / C23 0.1uF → GND

MAGNETOMETER

+3V0 → C20 0.1uF → GND

U3
QMC6309

VSS	A1	
SCL	A2	
SDA	B2	P1.11_I2CSDA_AIN4
VDD	B1	P1.12_I2CSCL_AIN5

U6
1532H

CLK_CTL	1	
NC		
VCC	4	+3V0
GND	2	GND
RF	3	CLK

TODO :
Button readable from Nordic MCU
Led actable from Nordic MCU
I2C address

Battery pin

TP6 → +BATT

USB-C

J1
GSB1C4621DS1HR

VBUS_A	A4_B9	VBUS_B	B4_A9
CC1	A5	SBU2	B8
DP1	A6	DN2	B7
DN1	A7	DP2	B6
SBU1	A8	CC2	B5
GND_A	A1_B12	GND_B	B1_A12
SHIELD	SH1		

USB_DRV → R1 1.5K 1%
VBUS → CC1 / USB_D_P / USB_D_N
VBUS → C17 10uF → GND
CC2 → R4 5.1K
CC1 → R5 5.1K
R4 / R5 → GND

+3V0 → C1 27P, C4 100N, C7 100N, C8 100N, C10 100N, C11 100N → GND

Sheet: /
File: NRF_ICM_QMC.kicad_sch

Title:

Size: A3	Date:	**Rev:**
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